

Technology Stack

Date	23 June2026
Team ID	
Project Name	Opticrop:Smart Agriculture Production Optimization
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your Architecture and provide a brief justification of your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS, Bootstrap	Used to create a responsive and user-friendly interface for entering agricultural data and displaying crop recommendations.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Provides fast processing of user inputs, integrates the machine learning model, and handles prediction requests efficiently.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MySQL	MySQL Stores crop-related information and supports efficient retrieval and management of agricultural data.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render / PythonAnywhere (or Local Host if not deployed)	Provides simple deployment and allows the application to be accessed online.

5	Version Control & CI/CD	GitHub	Enables version control, project backup, and seamless collaboration among team members.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib	Used for machine learning model development, data preprocessing, analysis, and visualization for crop prediction.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		